

1 From the electron-phonon interaction to the BCS ground state – details

1.1 Preface

These are supplementary notes to *Solid State Theory, Volume 2* by Gerd Czycholl. Since the derivations there are quite lengthy, only those steps are presented here that clarify what is done in the corresponding chapters. Numbers in parentheses such as (5.12) refer to equations of that book.

1.2 Attractive electron-electron interaction through the electron-phonon mechanism (Czycholl 5.2)

Two small observations.

- (5.12): use that S is linear in H_1 to discard $[[H_1, S], S]$.
- (5.15): use that $\langle m|H_1|m\rangle = 0$, since H_1 changes the number of phonons and thus $H_1|m\rangle \perp |m\rangle$.

The transformed Hamiltonian. Starting from (5.13) and inserting resolutions of the identity,

$$H_T = H_0 + \frac{i}{2}(H_1 S - S H_1) \quad (1)$$

$$= H_0 + \frac{i}{2} \sum_{\tilde{n}, n, m} \left(|\tilde{n}\rangle \langle \tilde{n}| H_1 |m\rangle \langle m| S |n\rangle \langle n| - |\tilde{n}\rangle \langle \tilde{n}| S |m\rangle \langle m| H_1 |n\rangle \langle n| \right) \quad (2)$$

$$\stackrel{(5.15)}{=} H_0 - \frac{1}{2} \sum_{\tilde{n}, n, m} \left(|\tilde{n}\rangle \langle \tilde{n}| H_1 |m\rangle \langle m| H_1 |n\rangle \langle n| \frac{1}{E_m - E_n} - |\tilde{n}\rangle \langle \tilde{n}| H_1 |m\rangle \langle m| H_1 |n\rangle \langle n| \frac{1}{E_{\tilde{n}} - E_m} \right), \quad (3)$$

where the matrix elements $\langle \tilde{n}|H_1|m\rangle$ and $\langle m|H_1|n\rangle$ are scalars and can be pulled out of the operator structure, which yields (5.16).

Restriction to a fixed phonon configuration. Since we want to derive a Hamiltonian for the electron-electron interaction only, we restrict $|\tilde{n}\rangle$ and $|n\rangle$ in (5.16) to states with the same phonon configuration, such that H_T does not mix states with different phonon configurations.

Let $|n\rangle$ be some many-particle state and assume that it contains one electron in the state $|k, \sigma\rangle$ and one in $|k', \sigma'\rangle$. The two elementary processes are

$$H_{1,1}|k, \sigma\rangle = \sqrt{n_q}|k + q, \sigma, n_q - 1\rangle, \quad \text{by destroying a phonon,} \quad (*_1)$$

$$H_{1,2}|k, \sigma\rangle = \sqrt{n_q + 1}|k - q, \sigma, n_q + 1\rangle, \quad \text{by creating a phonon.} \quad (*_2)$$

Since $|\tilde{n}\rangle$ and $|n\rangle$ are restricted to states with the same phonon number, the created (destroyed) phonon with momentum q has to be destroyed (created) again:

$$H_{1,1}\sqrt{n_q}|k+q, \sigma, n_q-1\rangle = n_q|k+q, \sigma; k'-q, \sigma'; n_q\rangle, \quad (4)$$

$$H_{1,1}\sqrt{n_q+1}|k-q, \sigma, n_q+1\rangle = (n_q+1)|k-q, \sigma; k'+q, \sigma'; n_q\rangle. \quad (5)$$

All other terms vanish, since their phonon configurations differ.

Energy denominators. Therefore we have

$$(*_1): \quad E_m - E_n = \varepsilon_{k+q} - \varepsilon_k - \hbar\omega_q, \quad E_{\tilde{n}} - E_m = \varepsilon_{k'-q} - \varepsilon_{k'} + \hbar\omega_q, \quad (5.19) \quad (6)$$

$$(*_2): \quad E_m - E_n = \varepsilon_{k-q} - \varepsilon_k + \hbar\omega_q, \quad E_{\tilde{n}} - E_m = \varepsilon_{k'+q} - \varepsilon_{k'} - \hbar\omega_q. \quad (5.20) \quad (7)$$

Inserting this into (5.16) gives

$$\begin{aligned} H_T = & -\frac{1}{2} \sum_{n, \tilde{n}} |\tilde{n}\rangle \langle n| \sum_m \sum_{\substack{k, k', q \\ \sigma, \sigma'}} \left(\langle \tilde{n}| M(-q) c_{k'-q, \sigma'}^\dagger c_{k', \sigma'} b_q^\dagger | m \rangle \langle m| M(q) c_{k+q, \sigma}^\dagger c_{k, \sigma} b_q | n \rangle \right. \\ & \times \left(\frac{1}{\varepsilon_{k+q} - \hbar\omega_q - \varepsilon_k} - \frac{1}{\varepsilon_{k'-q} - \varepsilon_{k'} + \hbar\omega_q} \right) \\ & + \langle \tilde{n}| M(q) c_{k'+q, \sigma'}^\dagger c_{k', \sigma'} b_q | m \rangle \langle m| M(-q) c_{k-q, \sigma}^\dagger c_{k, \sigma} b_q^\dagger | n \rangle \\ & \left. \times \left(\frac{1}{\varepsilon_{k-q} - \varepsilon_k + \hbar\omega_q} - \frac{1}{\varepsilon_{k'+q} - \varepsilon_{k'} - \hbar\omega_q} \right) \right), \end{aligned} \quad (8)$$

where the first summand is the $(*_1)$ -process and the second one the $(*_2)$ -process.

Using $M^*(q) = M(-q)$ and that all the above terms are independent of n , m , and $\tilde{n}$ yields (5.21). Using furthermore that

$$b_q^\dagger b_q | \dots n_q \dots \rangle = n_q | \dots n_q \dots \rangle, \quad b_q b_q^\dagger | \dots n_q \dots \rangle = (n_q + 1) | \dots n_q \dots \rangle \quad (9)$$

results in (5.21), i.e. $\hat{H}_T(b_q, b_q^\dagger, \dots) \rightsquigarrow \hat{H}_T(n_q, \dots)$ in the phonon number basis.

Combining the two processes. We have

$$\frac{1}{\varepsilon_{k+q} - \hbar\omega_q - \varepsilon_k} = \frac{(\varepsilon_{k+q} - \varepsilon_k) + \hbar\omega_q}{(\varepsilon_{k+q} - \varepsilon_k)^2 - \hbar^2\omega_q^2} \quad (10)$$

and, relabelling $k \leftrightarrow k'$ and $q \rightarrow -q$,

$$\sum_{k, k', q} \frac{1}{\varepsilon_{k'-q} - \varepsilon_{k'} + \hbar\omega_q} c_{k+q}^\dagger c_k c_{k'-q}^\dagger c_{k'} = \sum_{k, k', q} \frac{(\varepsilon_{k+q} - \varepsilon_k) - \hbar\omega_q}{(\varepsilon_{k+q} - \varepsilon_k)^2 - \hbar^2\omega_q^2} c_{k'-q}^\dagger c_{k'} c_{k+q}^\dagger c_k, \quad (11)$$

where

$$\sum_{k,k',q} c_{k'-q}^\dagger c_{k'} c_{k+q}^\dagger c_k = \sum_{k,k',q} c_{k'-q}^\dagger (\delta_{k',k+q} - c_{k+q}^\dagger c_{k'}) c_k \quad (12)$$

$$= \sum_{k,q} c_k^\dagger c_k - \sum_{k,k',q} c_{k+q}^\dagger c_{k'-q}^\dagger c_k c_{k'} \quad (13)$$

$$= \sum_{k,q} n_k - \sum_{k,k',q} c_{k+q}^\dagger (\delta_{k,k'-q} - c_k c_{k'-q}^\dagger) c_{k'} \quad (14)$$

$$= \underbrace{\sum_{k,q} n_k - \sum_{k,q} c_{k'}^\dagger c_{k'}}_{=0} + \sum_{k,k',q} c_{k+q}^\dagger c_k c_{k'-q}^\dagger c_{k'} \quad (15)$$

$$= \sum_{k,k',q} c_{k+q}^\dagger c_k c_{k'-q}^\dagger c_{k'}. \quad (16)$$

We know (Bruus & Flensberg, Sec. 3.8) that $\omega_{-q} = \omega_q$ and $M(-q) = M(q)$. Furthermore, since we are only interested in the electronic process, we can assume that there is no net phononic motion within the system and thus $n_q = n_{-q}$. Hence the first term of (5.21) equals the first term of (5.22), analogously for the second term, and similarly (5.22) $\rightsquigarrow$ (5.23).

1.3 Cooper pairs (Czycholl 5.3)

Setup. Simplify: look at the Fermi sea plus two electrons just above the Fermi energy, and only let them interact via H_T . The variational ansatz (5.26), for vanishing total pair momentum $K = k + k' \stackrel{!}{=} 0$, is

$$|\psi_{\sigma,\sigma'}\rangle = \sum_k a_{\sigma,\sigma'}(k) c_{k,\sigma}^\dagger c_{-k,\sigma'}^\dagger |\phi_0\rangle, \quad (17)$$

with the sum over k such that $E_F \leq \varepsilon_k, \varepsilon_{k'} \leq E_F + \hbar\omega_D$, cf. (5.27). The simplified interaction acts only outside the Fermi sea (*), otherwise $V = 0$:

$$H_T = -\frac{V}{2} \sum_{\sigma,\sigma'} \sum_{k,q} c_{k+q,\sigma}^\dagger c_{-k-q,\sigma'}^\dagger c_{-k,\sigma'} c_{k,\sigma}. \quad (18)$$

Action on the ansatz.

$$H_T|\psi_{\sigma_1\sigma_2}\rangle = -\frac{V}{2} \sum_{\sigma,\sigma'} \sum_{k,q,k'} c_{k+q,\sigma}^\dagger c_{-k-q,\sigma'}^\dagger c_{-k,\sigma'} c_{k,\sigma} a_{\sigma_1\sigma_2}(k') c_{k',\sigma_1}^\dagger c_{-k',\sigma_2}^\dagger |\phi_0\rangle \quad (19)$$

$$= -\frac{V}{2} \sum_{\sigma,\sigma'} \sum_{k,q,k'} a_{\sigma_1\sigma_2}(k') c_{k+q,\sigma}^\dagger c_{-k-q,\sigma'}^\dagger c_{-k,\sigma'} c_{k,\sigma} |k', \sigma_1; -k', \sigma_2\rangle \quad (20)$$

$$= -\frac{V}{2} \sum_{\sigma,\sigma'} \sum_{k,q,k'} a_{\sigma_1\sigma_2}(k') \left(\delta_{k,k'} \delta_{\sigma,\sigma_1} \delta_{\sigma',\sigma_2} |k+q, \sigma_1; -k-q, \sigma_2\rangle \right. \\ \left. + \delta_{k,-k'} \delta_{\sigma,\sigma_2} \delta_{\sigma',\sigma_1} | -k-q, \sigma_1; k+q, \sigma_2\rangle \right) \quad (21)$$

$$= -\frac{V}{2} \sum_{k,q} \left(a_{\sigma_1\sigma_2}(k) |k+q, \sigma_1; -k-q, \sigma_2\rangle + a_{\sigma_1\sigma_2}(-k) | -k-q, \sigma_1; k+q, \sigma_2\rangle \right), \quad (22)$$

where the projection onto $|k', \sigma_1; -k', \sigma_2\rangle$ is allowed because, by (*), only states outside the Fermi sea matter for the interaction. Taking the expectation value,

$$\langle\psi_{\sigma_1\sigma_2}|H_T|\psi_{\sigma_1\sigma_2}\rangle = -\frac{V}{2} \sum_{k,q,k'} a_{\sigma_1\sigma_2}^*(k') \left\{ a_{\sigma_1\sigma_2}(k) \langle k', \sigma_1; -k', \sigma_2 | k+q, \sigma_1; -k-q, \sigma_2 \rangle \right. \\ \left. + a_{\sigma_1\sigma_2}(-k) \langle k', \sigma_1; -k', \sigma_2 | -k-q, \sigma_1; k+q, \sigma_2 \rangle \right\} \quad (23)$$

$$= -\frac{V}{2} \sum_{k,q,k'} a_{\sigma_1\sigma_2}^*(k') \left\{ a_{\sigma_1\sigma_2}(k) \delta_{k',k+q} + a_{\sigma_1\sigma_2}(-k) \delta_{k',-k-q} \right\} \quad (24)$$

$$= -\frac{V}{2} \sum_{k,q} \left(a_{\sigma_1\sigma_2}^*(k+q) a_{\sigma_1\sigma_2}(k) + a_{\sigma_1\sigma_2}^*(-k-q) a_{\sigma_1\sigma_2}(-k) \right) \quad (25)$$

$$= -V \sum_{k,q} a_{\sigma_1\sigma_2}^*(k+q) a_{\sigma_1\sigma_2}(k), \quad (26)$$

where in the last step $k \rightarrow -k$, $q \rightarrow -q$ was substituted in the second term.

The kinetic part. Furthermore, by (5.17),

$$H_0 = \sum_{k,\sigma} \varepsilon_k n_{k,\sigma} + \sum_q \hbar \omega_q b_q^\dagger b_q. \quad (27)$$

Since the ansatz $|\psi_{\sigma\sigma'}\rangle$ is purely electronic, $\sum_q \hbar\omega_q b_q^\dagger b_q$ will not contribute. Splitting the electronic sum into states inside and outside the Fermi sea,

$$\langle\psi_{\sigma,\sigma'}|H_0|\psi_{\sigma,\sigma'}\rangle = \langle\psi_{\sigma\sigma'}|\left(\sum_{k,\tilde{\sigma}} \varepsilon_k n_{k,\tilde{\sigma}}\right)\left(\sum_{k'} a_{\sigma,\sigma'}(k') c_{k',\sigma}^\dagger c_{-k',\sigma'}^\dagger |\phi_0\rangle\right) \quad (28)$$

$$= \langle\psi_{\sigma\sigma'}|\left(\sum_{\substack{k \leq k_F \\ \tilde{\sigma}}} \varepsilon_k \hat{n}_{k,\tilde{\sigma}} + \sum_{\substack{k > k_F \\ \tilde{\sigma}}} \varepsilon_k \hat{n}_{k,\tilde{\sigma}}\right)\left(\sum_{k'} a_{\sigma,\sigma'}(k') c_{k',\sigma}^\dagger c_{-k',\sigma'}^\dagger |\phi_0\rangle\right) \quad (29)$$

$$= \langle\psi_{\sigma\sigma'}|\sum_{\substack{k \leq k_F \\ \tilde{\sigma}}} \varepsilon_k n_{k,\tilde{\sigma}} |\psi_{\sigma\sigma'}\rangle + \sum_{k,k'} a_{\sigma,\sigma'}(k') (\varepsilon_{k'} \delta_{k,k'} \delta_{\sigma,\tilde{\sigma}} + \varepsilon_{k'} \delta_{k,-k'} \delta_{\sigma',\tilde{\sigma}}) c_{k',\sigma}^\dagger c_{-k',\sigma'}^\dagger |\phi_0\rangle \quad (30)$$

$$= \left(\langle\phi_0|\sum_k c_{-k,\sigma'} c_{k,\sigma} a_{\sigma,\sigma'}^*(k)\right) \left(\sum_{\substack{k \leq k_F \\ \tilde{\sigma}}} \varepsilon_k n_{k,\tilde{\sigma}} |\psi_{\sigma\sigma'}\rangle + \sum_{k'} a_{\sigma,\sigma'}(k') (\varepsilon_{k'} + \varepsilon_{-k'}) c_{k',\sigma}^\dagger c_{-k',\sigma'}^\dagger |\phi_0\rangle\right) \quad (31)$$

$$= \sum_k |a_{\sigma,\sigma'}(k)|^2 \left((\varepsilon_k + \varepsilon_{-k}) + \sum_{\substack{k \leq k_F \\ \tilde{\sigma}}} \varepsilon_k n_{k,\tilde{\sigma}} \right) \quad (32)$$

$$= \sum_k |a_{\sigma,\sigma'}(k)|^2 (\varepsilon_k + \varepsilon_{-k}) + c \sum_k |a_{\sigma\sigma'}(k)|^2, \quad (33)$$

where c collects the (constant) energy of the filled Fermi sea. From $\langle\psi_{\sigma\sigma'}|\psi_{\sigma\sigma'}\rangle \stackrel{!}{=} 1$ it follows that $\sum_k |a_{\sigma\sigma'}(k)|^2 \stackrel{!}{=} 1$, cf. (5.29), and therefore

$$H_0 = \sum_k |a_{\sigma,\sigma'}(k)|^2 (\varepsilon_k + \varepsilon_{-k}), \quad \implies \quad H = H_0 + H_T = (5.29). \quad (34)$$

The step $\varepsilon_k = \varepsilon_{-k}$ needed for (5.30) is true due to Kramers' theorem; see e.g. Sec. 5.3 of Czycholl I.

When does the gap equation degenerate? For (5.32) it was assumed that $C \equiv \sum_k a_{\sigma\sigma'}(k) \neq 0$. Let us examine when C is actually zero. We have

$$a_{\sigma\sigma'}(k) = \frac{V \sum_{k'} a_{\sigma\sigma'}(k')}{2\varepsilon_k - \lambda}, \quad (5.31) \quad (35)$$

where $\varepsilon_k = \varepsilon_{-k}$ by Kramers' theorem, and hence

$$a_{\sigma\sigma'}(-k) = a_{\sigma\sigma'}(k). \quad (*)$$

Moreover, for equal spins,

$$|\psi_{\sigma\sigma}\rangle = \sum_k a_{\sigma,\sigma}(k) c_{k,\sigma}^\dagger c_{-k,\sigma}^\dagger |\phi_0\rangle \quad (36)$$

$$= \sum_k a_{\sigma,\sigma}(k) (\delta_{k,-k} - c_{-k,\sigma}^\dagger c_{k,\sigma}^\dagger) |\phi_0\rangle \quad (37)$$

$$= - \sum_k a_{\sigma,\sigma}(k) c_{-k,\sigma}^\dagger c_{k,\sigma}^\dagger |\phi_0\rangle \quad (38)$$

$$= - \sum_k a_{\sigma,\sigma}(-k) c_{k,\sigma}^\dagger c_{-k,\sigma}^\dagger |\phi_0\rangle \quad (39)$$

$$= - \sum_k a_{\sigma,\sigma}(k) c_{k,\sigma}^\dagger c_{-k,\sigma}^\dagger |\phi_0\rangle = -|\psi_{\sigma\sigma}\rangle. \quad (40)$$

Here $\delta_{k,-k}$ drops out because $\sum_k$ does not include momenta inside the Fermi sea and thus $k \neq 0$; in the fourth line $k \rightarrow -k$ was substituted, and in the fifth line (*) was used. Consequently

$$\langle \psi_{\sigma\sigma} | \psi_{\sigma\sigma} \rangle = -\langle \psi_{\sigma\sigma} | \psi_{\sigma\sigma} \rangle \implies \langle \psi_{\sigma\sigma} | \psi_{\sigma\sigma} \rangle = 0 \implies \sum_k |a_{\sigma\sigma}(k)|^2 = 0 \implies a_{\sigma\sigma}(k) = 0, \quad (41)$$

so $\sum_k a_{\sigma\sigma}(k) = 0$. Thus $\sigma = \sigma'$ yields $C = 0$, cf. (5.35), and Cooper pairs consist of electrons with opposite spins.

Note that if $\psi(r_1 - r_2)$ is antisymmetric, then $\psi(0) = 0$ and $C = 0$ as well, so the derivation of (5.35) is different in that case.

Position representation. Project the state $|\psi_{\sigma,\sigma'}\rangle = \sum_k a_{\sigma,\sigma'}(k) c_{k,\sigma}^\dagger c_{-k,\sigma'}^\dagger |\phi_0\rangle$ onto the position basis. Since H acts trivially on the spin space ($\sigma \rightarrow \sigma, \sigma' \rightarrow \sigma'$), one can factorize the wave function:

$$\langle \mathbf{r}_1, \mathbf{r}_2 | \psi_{\sigma,\sigma'} \rangle = \psi_{\sigma,\sigma'}(\mathbf{r}_1, \mathbf{r}_2) = \psi(\mathbf{r}_1, \mathbf{r}_2) \chi(\sigma, \sigma'), \quad (5.37) \quad (42)$$

with

$$\psi(\mathbf{r}_1, \mathbf{r}_2) = \frac{1}{V} \sum_k a(k) e^{i\mathbf{k}\cdot\mathbf{r}_1} e^{-i\mathbf{k}\cdot\mathbf{r}_2} = \psi(\mathbf{r}_1 - \mathbf{r}_2), \quad (5.36) \quad (43)$$

using $\langle \mathbf{r} | \mathbf{k} \rangle = \frac{1}{\sqrt{V}} e^{i\mathbf{k}\cdot\mathbf{r}}$.

1.4 BCS theory (Czychołł 5.4)

Mean-field recap. Write $\hat{n}_\alpha = \langle n_\alpha \rangle + \delta\hat{n}_\alpha$ and assume that $\delta\hat{n}_\alpha$ is small, such that second-order fluctuations can be dropped:

$$\hat{n}_\alpha \hat{n}_\beta = (\langle \hat{n}_\alpha \rangle + \delta\hat{n}_\alpha) (\langle \hat{n}_\beta \rangle + \delta\hat{n}_\beta) \quad (44)$$

$$\stackrel{\delta\hat{n}_\alpha \delta\hat{n}_\beta \approx 0}{=} \langle \hat{n}_\alpha \rangle \langle \hat{n}_\beta \rangle + \langle \hat{n}_\alpha \rangle (\hat{n}_\beta - \langle \hat{n}_\beta \rangle) + \langle \hat{n}_\beta \rangle (\hat{n}_\alpha - \langle \hat{n}_\alpha \rangle) \quad (45)$$

$$= \langle \hat{n}_\alpha \rangle \hat{n}_\beta + \langle \hat{n}_\beta \rangle \hat{n}_\alpha - \langle \hat{n}_\alpha \rangle \langle \hat{n}_\beta \rangle. \quad (46)$$

Now apply this to $c_{-k\downarrow}^{(\dagger)} c_{k\uparrow}^{(\dagger)}$ instead of $\hat{n}_\alpha$ to derive a mean-field version of the BCS Hamiltonian, which gives (5.40).

Nambu form of the effective Hamiltonian. Rewrite H_{eff} as (following Becca & Sorella, and the Superconductivity IMPRS UFAST 2017 course of Michael Sentef)

$$\tilde{H}_{\text{eff}} = \sum_k \begin{pmatrix} c_{k\uparrow}^\dagger & c_{-k\downarrow} \end{pmatrix} \begin{pmatrix} \varepsilon_k & -\Delta \\ -\Delta^* & -\varepsilon_k \end{pmatrix} \begin{pmatrix} c_{k\uparrow} \\ c_{-k\downarrow}^\dagger \end{pmatrix} \quad (47)$$

$$= \sum_k \begin{pmatrix} c_{k\uparrow}^\dagger & c_{-k\downarrow} \end{pmatrix} \begin{pmatrix} \varepsilon_k c_{k\uparrow} - \Delta c_{-k\downarrow}^\dagger \\ -\Delta^* c_{k\uparrow} - \varepsilon_k c_{-k\downarrow}^\dagger \end{pmatrix} \quad (48)$$

$$= \sum_k \varepsilon_k c_{k\uparrow}^\dagger c_{k\uparrow} - \Delta c_{k\uparrow}^\dagger c_{-k\downarrow}^\dagger - \Delta^* c_{-k\downarrow} c_{k\uparrow} - \varepsilon_k \underbrace{c_{-k\downarrow} c_{-k\downarrow}^\dagger}_{= 1 - c_{-k\downarrow}^\dagger c_{-k\downarrow}} \quad (49)$$

$$= \sum_{k,\sigma} \varepsilon_k c_{k\sigma}^\dagger c_{k\sigma} - \Delta^* \sum_k c_{-k\downarrow} c_{k\uparrow} - \Delta \sum_k c_{k\uparrow}^\dagger c_{-k\downarrow}^\dagger - \sum_k \varepsilon_k \quad (50)$$

$$= H_{\text{eff}} - \underbrace{\frac{|\Delta|^2}{V} - \sum_k \varepsilon_k}_{\text{const.}} \quad (5.41) \quad (51)$$

Set $M \equiv \begin{pmatrix} \varepsilon_k & -\Delta \\ -\Delta^* & -\varepsilon_k \end{pmatrix}$ and diagonalize it, such that H_{eff} becomes diagonal — this is a unitary transformation.

Diagonalizing the 2×2 matrix M . Eigenvalues:

$$\det \begin{pmatrix} \varepsilon_k - \lambda & -\Delta \\ -\Delta^* & -\varepsilon_k - \lambda \end{pmatrix} = -(\varepsilon_k - \lambda)(\varepsilon_k + \lambda) - |\Delta|^2 \quad (52)$$

$$= -(\lambda^2 - \varepsilon_k^2 - |\Delta|^2) \stackrel{!}{=} 0 \quad (53)$$

$$\implies \lambda_{1,2} = \pm \sqrt{\varepsilon_k^2 + |\Delta|^2} \equiv E_k^\pm \in \mathbb{R}. \quad (*_1)$$

Eigenvectors: from

$$\begin{pmatrix} \varepsilon_k - E_k^\pm & -\Delta \\ -\Delta^* & -\varepsilon_k - E_k^\pm \end{pmatrix} \begin{pmatrix} x_\pm \\ y_\pm \end{pmatrix} \stackrel{!}{=} 0 \quad (54)$$

the first row (I) gives

$$(\varepsilon_k - E_k^\pm) x_\pm - \Delta y_\pm = 0 \iff y_\pm = \frac{\varepsilon_k - E_k^\pm}{\Delta} x_\pm, \quad (\text{I}') \quad (55)$$

while the second row (II) gives, using (I') and $(E_k^\pm)^2 = \varepsilon_k^2 + |\Delta|^2$,

$$\Delta^* x_\pm + (\varepsilon_k + E_k^\pm) y_\pm \stackrel{(\text{I}')}{=} \left(\Delta^* + \frac{\varepsilon_k^2 - (E_k^\pm)^2}{\Delta} \right) x_\pm = 0, \quad (55)$$

i.e. it is satisfied identically. Hence $x_\pm$ can be chosen arbitrarily. $(\odot)$

Normalization. Requiring $|x_{\pm}|^2 + |y_{\pm}|^2 \stackrel{!}{=} 1$ and using (I'),

$$|x_{\pm}|^2 \left(1 + \frac{(\varepsilon_k - E_k^{\pm})^2}{|\Delta|^2} \right) = |x_{\pm}|^2 \frac{|\Delta|^2 + (\varepsilon_k - E_k^{\pm})^2}{|\Delta|^2} \stackrel{!}{=} 1, \quad (56)$$

where

$$(\varepsilon_k - E_k^{\pm})^2 = \varepsilon_k^2 - 2\varepsilon_k E_k^{\pm} + (E_k^{\pm})^2 \quad (57)$$

$$= \varepsilon_k^2 - 2\varepsilon_k E_k^{\pm} + \varepsilon_k^2 + |\Delta|^2 \quad (58)$$

$$= 2\varepsilon_k^2 - 2\varepsilon_k E_k^{\pm} + |\Delta|^2. \quad (59)$$

Therefore

$$|x_{\pm}|^2 = \frac{|\Delta|^2}{2(\varepsilon_k^2 - \varepsilon_k E_k^{\pm} + |\Delta|^2)} \stackrel{(*)}{=} \frac{1}{2} \frac{|\Delta|^2}{(E_k^{\pm})^2 - \varepsilon_k E_k^{\pm}} \stackrel{(*)}{=} \frac{1}{2} \frac{(E_k^{\pm})^2 - \varepsilon_k^2}{E_k^{\pm}(E_k^{\pm} - \varepsilon_k)} \quad (60)$$

$$= \frac{1}{2} \frac{E_k^{\pm} + \varepsilon_k}{E_k^{\pm}} = \frac{1}{2} \left(1 + \frac{\varepsilon_k}{E_k^{\pm}} \right) = \frac{1}{2} \left(1 \pm \frac{\varepsilon_k}{E_k} \right), \quad (61)$$

$$|y_{\pm}|^2 = 1 - |x_{\pm}|^2 = \frac{1}{2} \left(1 - \frac{\varepsilon_k}{E_k^{\pm}} \right) = \frac{1}{2} \left(1 \mp \frac{\varepsilon_k}{E_k} \right). \quad (A)$$

Fixing the free phase. Choose $x_{\pm}$ to be some real value, which is allowed by ($\odot$). Then (A) gives $x_{\pm}^2 = y_{\mp}^2$, i.e. $x_{\pm} = \pm y_{\mp}$, which is still too many solutions. Set

$$x_+ = y_- \equiv u_k, \quad x_- = -y_+ \equiv v_k. \quad (62)$$

Since $M_k^{\dagger} = M_k$, we know that M_k can be diagonalized by a unitary U_k :

$$M = \begin{pmatrix} \varepsilon_k & -\Delta \\ -\Delta^* & -\varepsilon_k \end{pmatrix} = U_k^{\dagger} \begin{pmatrix} E_k^+ & 0 \\ 0 & E_k^- \end{pmatrix} U_k, \quad U_k = \begin{pmatrix} x^+ & x^- \\ y^+ & y^- \end{pmatrix} = \begin{pmatrix} u_k & v_k \\ -v_k & u_k \end{pmatrix}, \quad (63)$$

which is obviously unitary.

The Bogoliubov transformation. Inserting this into the Hamiltonian yields

$$\tilde{H}_{\text{eff}} = \sum_k \begin{pmatrix} c_{k\uparrow}^{\dagger} & c_{-k\downarrow} \end{pmatrix} U_k U_k^{\dagger} \begin{pmatrix} \varepsilon_k & -\Delta \\ -\Delta^* & -\varepsilon_k \end{pmatrix} U_k U_k^{\dagger} \begin{pmatrix} c_{k\uparrow} \\ c_{-k\downarrow}^{\dagger} \end{pmatrix} - \sum_k \varepsilon_k \quad (64)$$

$$= \sum_k \begin{pmatrix} \alpha_k^{\dagger} & \beta_k \end{pmatrix} \begin{pmatrix} E_k^+ & 0 \\ 0 & E_k^- \end{pmatrix} \begin{pmatrix} \alpha_k \\ \beta_k^{\dagger} \end{pmatrix} - \sum_k \varepsilon_k \quad (65)$$

$$= \sum_k E_k^+ \alpha_k^{\dagger} \alpha_k + E_k^- \beta_k \beta_k^{\dagger} - \sum_k \varepsilon_k \quad (66)$$

$$= \sum_k E_k^+ (\alpha_k^{\dagger} \alpha_k + \beta_k^{\dagger} \beta_k) - \underbrace{\sum_k (\varepsilon_k + E_k^+)}_{\text{const.}}, \quad \sim (5.46) \quad (67)$$

where $\{\beta_k, \beta_k^\dagger\} = 1$ and $E_k^- = -E_k^+$ were used, and with $\begin{pmatrix} \alpha_k \\ \beta_k^\dagger \end{pmatrix} = U_k^\dagger \begin{pmatrix} c_{k\uparrow} \\ c_{-k\downarrow}^\dagger \end{pmatrix}$. Here U_k is known as the Bogoliubov transformation, with the corresponding Bogoliubov quasiparticles

$$\alpha_k = u_k c_{k\uparrow} - v_k c_{-k\downarrow}^\dagger, \quad (68)$$

$$\beta_k = u_k c_{-k\downarrow} + v_k c_{k\uparrow}^\dagger, \quad (69)$$

which reproduces (5.43)–(5.47). They obey the usual fermionic anticommutation relations, as can be seen as follows (for arbitrary unitary transformations).

Nambu spinors and the anticommutation relations. Define

$$\psi_k = \begin{pmatrix} c_{k\uparrow} \\ c_{-k\downarrow}^\dagger \end{pmatrix}, \quad \psi_k^\dagger = \begin{pmatrix} c_{k\uparrow}^\dagger & c_{-k\downarrow} \end{pmatrix} \quad (\text{Nambu spinors}), \quad (70)$$

with $\{\psi_{k,i}, \psi_{k,j}^\dagger\} = \psi_{k,i} \psi_{k,j}^\dagger + \psi_{k,j}^\dagger \psi_{k,i} = \delta_{ij}$. The Bogoliubov transformation can then be written as

$$\Gamma_k \equiv \begin{pmatrix} \alpha_k \\ \beta_k^\dagger \end{pmatrix} = U^\dagger \psi_k, \quad \Gamma_k^\dagger \equiv \psi_k^\dagger U, \quad (71)$$

so that, using bilinearity of $\{\cdot, \cdot\}$ and that the $U_{\alpha\beta}^{(\dagger)}$ are numbers,

$$\{\Gamma_{k,i}, \Gamma_{k,j}^\dagger\} = \left\{ \sum_\ell U_{i,\ell}^\dagger \psi_{k,\ell}, \sum_m \psi_{k,m}^\dagger U_{m,j} \right\} \quad (72)$$

$$= \sum_{\ell,m} U_{i,\ell}^\dagger \{\psi_{k,\ell}, \psi_{k,m}^\dagger\} U_{m,j} \quad (73)$$

$$= \sum_\ell U_{i,\ell}^\dagger U_{\ell,j} = (U^\dagger U)_{ij} = \delta_{ij}. \quad (74)$$

Since U is unitary, α_k and β_k are fermionic operators.

Coherence factors. With (5.45),

$$(u_k v_k)^2 = \frac{1}{4} \left(1 - \frac{\varepsilon_k^2}{(E_k^+)^2} \right) = \frac{1}{4} \left(\frac{\varepsilon_k^2 + |\Delta|^2 - \varepsilon_k^2}{\varepsilon_k^2 + |\Delta|^2} \right) \quad (75)$$

$$= \frac{1}{4} \frac{|\Delta|^2}{(E_k^+)^2} \implies u_k v_k = \pm \frac{1}{2} \frac{|\Delta|}{E_k^+}, \quad (76)$$

from which the rest follows, i.e. (5.51), where the negative solution leads to the contradiction $1 = \text{negative value}$.

The BCS ground state. Since α_k is fermionic, $n_{\alpha,k}^2 \equiv (\alpha_k^\dagger \alpha_k)^2 = \alpha_k^\dagger \alpha_k$, and thus any eigenvalue λ of $n_{\alpha,k}$ must satisfy $\lambda^2 = \lambda$, i.e. $\lambda \in \{0, 1\}$ (and the same for $n_{\beta,k}$). Since $E_k^+ \geq 0$, it follows that if $\alpha_k |\psi_{\text{BCS}}\rangle = \beta_k |\psi_{\text{BCS}}\rangle = 0$, then $|\psi_{\text{BCS}}\rangle$ is the ground state of $H_{\text{eff}} = \sum_k E_k^+ (\alpha_k^\dagger \alpha_k + \beta_k^\dagger \beta_k)$.

The pairing ansatz. Following Becca & Sorella (1.91),

$$|\psi_{\text{BCS}}\rangle = \exp\left(\sum_k f_k c_{k\uparrow}^\dagger c_{-k\downarrow}^\dagger\right)|0\rangle \quad (77)$$

$$= \prod_k \exp(f_k c_{k\uparrow}^\dagger c_{-k\downarrow}^\dagger)|0\rangle \quad (78)$$

$$= \prod_k \left[\sum_{n=0}^{\infty} \frac{f_k^n}{n!} (c_{k\uparrow}^\dagger)^n (c_{-k\downarrow}^\dagger)^n \right] |0\rangle, \quad \text{where } (c_\alpha^\dagger)^2 |0\rangle = 0, \quad (79)$$

$$= \prod_k (1 + f_k c_{k\uparrow}^\dagger c_{-k\downarrow}^\dagger) |0\rangle \quad (80)$$

$$= \bigotimes_k \left(|0\rangle_k + f_k c_{k\uparrow}^\dagger c_{-k\downarrow}^\dagger |0\rangle_k \right), \quad (81)$$

with the one-momentum vacuum $|0\rangle_k$ defined by $|0\rangle = \bigotimes_k |0\rangle_k$ and $c_{k\uparrow}^\dagger |0\rangle_k = c_{-k\downarrow}^\dagger |0\rangle_k = 0$. Thus this ansatz is a pure product state, $|\psi_{\text{BCS}}\rangle = \bigotimes_k |\psi_{\text{BCS}}\rangle_k$, and therefore

$$\alpha_k |\psi_{\text{BCS}}\rangle = \alpha_k \bigotimes_{k'} |\psi_{\text{BCS}}\rangle_{k'} = |\psi\rangle_{k_1} \otimes \cdots \otimes \alpha_k |\psi_{\text{BCS}}\rangle_k \otimes \cdots \quad (82)$$

so that only a single momentum sector has to be examined:

$$\alpha_k |\psi_{\text{BCS}}\rangle_k = (u_k c_{k\uparrow} - v_k c_{-k\downarrow}^\dagger) \left(|0\rangle_k + f_k c_{k\uparrow}^\dagger c_{-k\downarrow}^\dagger |0\rangle_k \right) \quad (83)$$

$$= (u_k f_k c_{k\uparrow} c_{k\uparrow}^\dagger c_{-k\downarrow}^\dagger - v_k c_{-k\downarrow}^\dagger) |0\rangle_k \stackrel{!}{=} 0 \quad (84)$$

$$\Longleftrightarrow v_k c_{-k\downarrow}^\dagger |0\rangle_k = u_k f_k (1 - c_{k\uparrow}^\dagger c_{k\uparrow}) c_{-k\downarrow}^\dagger |0\rangle_k \quad (85)$$

$$= u_k f_k \left(c_{-k\downarrow}^\dagger + c_{k\uparrow}^\dagger \underbrace{c_{-k\downarrow}^\dagger c_{k\uparrow}}_{=0} \right) |0\rangle_k, \quad (86)$$

where $c_\alpha |0\rangle = 0$ and $c_\alpha^\dagger c_\alpha^\dagger |0\rangle = 0$ were used. Since $u_k \neq 0$ this gives $f_k = v_k/u_k$, and similarly

$$\beta_k |\psi_{\text{BCS}}\rangle_k = (u_k f_k c_{-k\downarrow} c_{k\uparrow}^\dagger c_{-k\downarrow}^\dagger + v_k c_{k\uparrow}^\dagger) |0\rangle_k \stackrel{!}{=} 0 \quad (87)$$

$$\Longleftrightarrow v_k c_{k\uparrow}^\dagger |0\rangle_k = u_k f_k c_{k\uparrow}^\dagger (1 - c_{-k\downarrow}^\dagger c_{-k\downarrow}) |0\rangle_k, \quad (88)$$

again yielding $f_k = v_k/u_k$.

Therefore $|\psi_{\text{BCS}}\rangle$ is indeed the ground state of the mean-field Hamiltonian, and it behaves like the vacuum of the Bogoliubons. To avoid problems due to $u_k = 0$, one often writes instead

$$|\psi_{\text{BCS}}\rangle = \prod_k (u_k + v_k c_{k\uparrow}^\dagger c_{-k\downarrow}^\dagger) |0\rangle. \quad (89)$$

1.5 References

- G. Czycholl, *Solid State Theory*, Volume 2 (and Sec. 5.3 of Volume 1 for Kramers' theorem).

- H. Bruus & K. Flensberg, *Many-Body Quantum Theory in Condensed Matter Physics*, Sec. 3.8.
- F. Becca & S. Sorella, *Quantum Monte Carlo Approaches for Correlated Systems*, Eq. (1.91).
- M. Sentef, *Superconductivity*, IMPRS UFAST 2017 course.